

Longitudinal Analysis of Multimodal Pain Management Prescribing Among Spine Surgeons

United States, 2014–2023
Medicare Part B & Part D Claims Data

Data Source: Centers for Medicare & Medicaid Services (CMS)
Medicare Provider Utilization and Payment Data
Accessed via CMS data.cms.gov API

Key Findings

Study Population: 6,810 unique procedure-confirmed spine surgeons identified across 2014–2023 (neurosurgery + orthopedic surgery performing spine CPT codes). ~4,198 spine surgeons per year in 2014 declining to ~3,925 in 2023. All 10 years of data (2014–2023) included.

Opioid Prescribing (–51.7%, $p<0.001$): Total opioid claims fell from 707,906 (2014) to 341,801 (2023). Claims per provider declined from 190.3 to 135.6 (–28.7%, $p<0.001$). Average days supply per opioid claim decreased from 15.4 to 11.1 days (–28.3%, $p<0.001$). Total opioid drug cost dropped from \$18.8M to \$4.4M (–76.8%, $p<0.001$). Cost per opioid claim fell from \$26.52 to \$12.73 (–52.0%).

Gabapentinoid Prescribing (+39.3%, $p<0.001$): Total claims rose from 151,485 to 211,067. Claims per provider increased from 59.7 to 86.4 (+44.7%, $p<0.001$). Despite volume increase, cost per claim fell from \$40.58 to \$15.10 (–62.8%) due to generic availability.

Muscle Relaxant Prescribing (+59.3%, $p<0.001$): Total claims rose from 113,116 to 180,212. Claims per provider increased from 58.2 to 81.6 (+40.3%, $p<0.001$). Cost per claim fell from \$20.67 to \$10.47 (–49.3%).

NSAID Prescribing (+22.8%, $p<0.05$): Total claims rose from 135,554 to 166,415. Claims per provider increased from 81.6 to 103.4 (+26.7%, $p<0.001$). Cost per claim fell from \$43.75 to \$15.16 (–65.3%).

Proportion Shift: Opioids comprised 63.9% of all pain management claims in 2014 but only 38.0% by 2023. Non-opioid modalities (gabapentinoids, muscle relaxants, NSAIDs) collectively grew from 36.1% to 62.0%. Total non-opioid claims surpassed opioid claims in 2019.

Table 1: Total Claims by Drug Category and Year

Year	Opioid	Gabapentinoid	Muscle Relaxant	NSAID	Total
2014	707,906	151,485	113,116	135,554	1,108,061
2015	632,887	162,994	106,785	140,799	1,043,465
2016	620,368	179,237	104,936	145,562	1,050,103
2017	572,680	188,845	126,679	146,272	1,034,476
2018	511,154	192,547	127,850	145,705	977,256
2019	449,991	187,295	137,559	150,059	924,904
2020	396,114	190,944	135,308	136,056	858,422
2021	374,916	207,413	154,581	146,589	883,499
2022	360,937	213,107	165,541	157,433	897,018
2023	341,801	211,067	180,212	166,415	899,495

Table 2: Mean Claims Per Provider by Drug Category and Year

Year	Opioid	Gabapentinoid	Muscle Relaxant	NSAID
2014	190.3	59.7	58.2	81.6
2015	174.1	63.4	55.4	82.3
2016	170.4	67.7	57.2	82.4
2017	163.4	69.4	58.0	83.1
2018	150.9	71.8	60.8	86.1
2019	139.4	71.0	65.0	88.1
2020	139.1	76.5	67.2	87.6
2021	134.9	83.0	72.7	92.3
2022	135.3	85.4	76.4	98.8
2023	135.6	86.4	81.6	103.4

Table 3: Average Days Supply Per Claim

Year	Opioid	Gabapentinoid	Muscle Relaxant	NSAID
2014	15.4	33.7	23.4	32.9
2015	16.5	34.0	24.1	33.5
2016	16.6	34.3	24.6	33.9
2017	16.4	34.6	24.5	34.5
2018	15.0	35.0	24.3	36.0
2019	12.8	35.6	23.6	36.3

2020	12.4	34.8	23.2	35.9
2021	11.7	34.4	22.6	35.3
2022	11.4	34.1	22.4	35.2
2023	11.1	34.2	22.3	35.2

Cost Analysis

Table 4: Total Drug Cost by Category and Year (\$ Millions)

Year	Opioid	Gabapentinoid	Muscle Relax.	NSAID	Total
2014	\$18.8M	\$6.1M	\$2.3M	\$5.9M	\$33.2M
2015	\$18.2M	\$6.5M	\$2.1M	\$4.3M	\$31.0M
2016	\$17.1M	\$6.7M	\$1.5M	\$4.2M	\$29.5M
2017	\$12.5M	\$7.1M	\$1.8M	\$3.9M	\$25.3M
2018	\$11.0M	\$8.3M	\$1.7M	\$3.2M	\$24.2M
2019	\$7.6M	\$5.6M	\$1.8M	\$2.7M	\$17.7M
2020	\$6.0M	\$3.0M	\$1.8M	\$2.5M	\$13.3M
2021	\$5.3M	\$3.2M	\$1.9M	\$2.4M	\$12.9M
2022	\$4.9M	\$3.3M	\$2.0M	\$2.5M	\$12.7M
2023	\$4.4M	\$3.2M	\$1.9M	\$2.5M	\$12.0M

Table 5: Cost Per Claim by Drug Category and Year

Year	Opioid	Gabapentinoid	Muscle Relax.	NSAID
2014	\$26.52	\$40.58	\$20.67	\$43.75
2015	\$28.75	\$39.59	\$19.32	\$30.56
2016	\$27.61	\$37.39	\$14.14	\$28.99
2017	\$21.84	\$37.61	\$13.90	\$26.86
2018	\$21.46	\$43.21	\$13.57	\$21.90
2019	\$16.86	\$29.97	\$12.89	\$17.90
2020	\$15.25	\$15.91	\$13.03	\$18.38
2021	\$14.26	\$15.55	\$12.45	\$16.56
2022	\$13.50	\$15.68	\$12.12	\$15.71
2023	\$12.73	\$15.10	\$10.47	\$15.16

Cost Highlights: Total pain management drug spending by spine surgeons fell from \$33.2M (2014) to \$12.0M (2023), a 63.9% decline. Opioid spending drove most of this reduction: opioid costs fell 76.8% (\$18.8M → \$4.4M). Opioid's share of total drug spending fell from 56.6% to 36.4%. Gabapentinoid cost per claim dropped 62.8% (\$40.58 → \$15.10), reflecting generic gabapentin/pregabalin availability. NSAID cost per claim fell 65.3% (\$43.75 → \$15.16). Despite rising claim volumes for all non-opioid categories, total spending still decreased due to dramatic per-unit cost reductions.

Specialty Analysis: Neurosurgery vs. Orthopedic Surgery

Orthopedic spine surgeons consistently prescribed at higher volumes per provider than neurosurgeons across all drug categories. Both specialties showed similar opioid reduction rates, but orthopedic surgeons adopted non-opioid alternatives at higher absolute rates.

Table 6: Opioid Claims Per Provider by Specialty

Year	Neurosurgery	Orthopedic Surg.	Neuro N	Ortho N
2014	146.5	241.3	2,002	1,718
2015	132.2	221.1	1,921	1,714
2016	129.5	215.6	1,911	1,730
2017	120.4	209.0	1,812	1,696
2018	117.0	185.9	1,721	1,666
2019	110.1	168.2	1,596	1,631
2020	107.6	168.8	1,375	1,470
2021	106.4	160.9	1,326	1,453
2022	103.9	162.5	1,235	1,432
2023	106.6	161.1	1,176	1,344

Table 7: Gabapentinoid Claims Per Provider by Specialty

Year	Neurosurgery	Orthopedic Surg.	Neuro N	Ortho N
2014	48.8	71.2	1,302	1,235
2015	49.9	78.0	1,327	1,242
2016	53.7	81.5	1,318	1,331
2017	53.3	87.3	1,322	1,356
2018	53.6	89.2	1,307	1,374
2019	52.4	87.6	1,245	1,393
2020	55.5	94.8	1,147	1,342
2021	60.5	100.9	1,157	1,361
2022	59.4	106.6	1,121	1,374
2023	59.7	108.1	1,097	1,346

Specialty Highlights: Neurosurgeons reduced opioid claims/provider from 146.5 to 106.6 (−27.2%), while orthopedic surgeons reduced from 241.3 to 161.1 (−33.2%). Orthopedic surgeons showed greater absolute adoption of gabapentinoids (+51.8%, from 71.2 to 108.1 claims/provider) compared to neurosurgeons (+22.3%, 48.8 to 59.7). The number of neurosurgery opioid prescribers fell 41.3% (2,002 → 1,176), while orthopedic opioid prescribers fell 21.8% (1,718 → 1,344).

Figure 1: Total Pain Management Claims by Drug Category

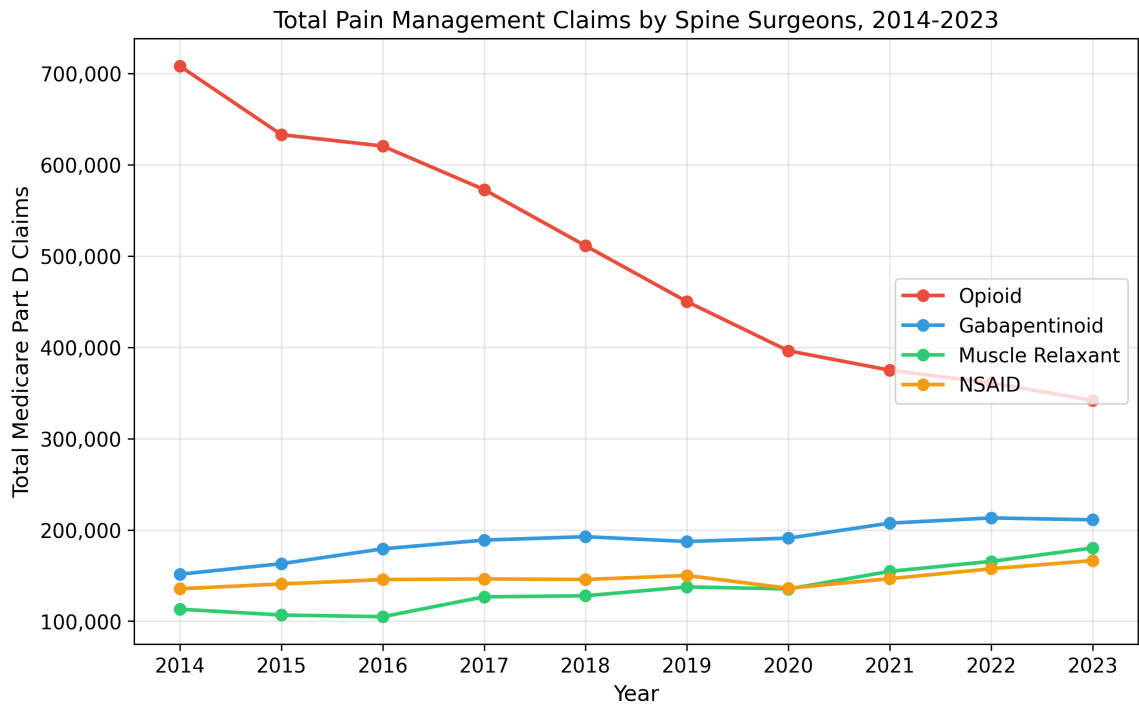


Figure 2: Mean Claims Per Spine Surgeon Prescriber

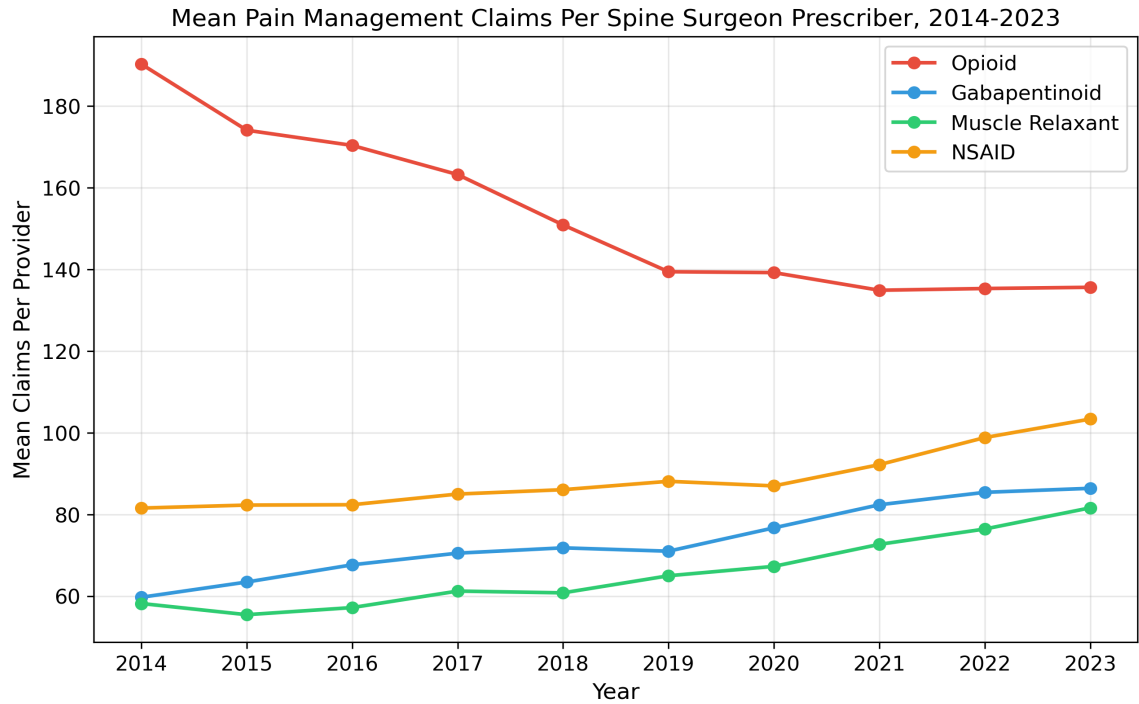


Figure 3: Number of Spine Surgeon Prescribers by Category

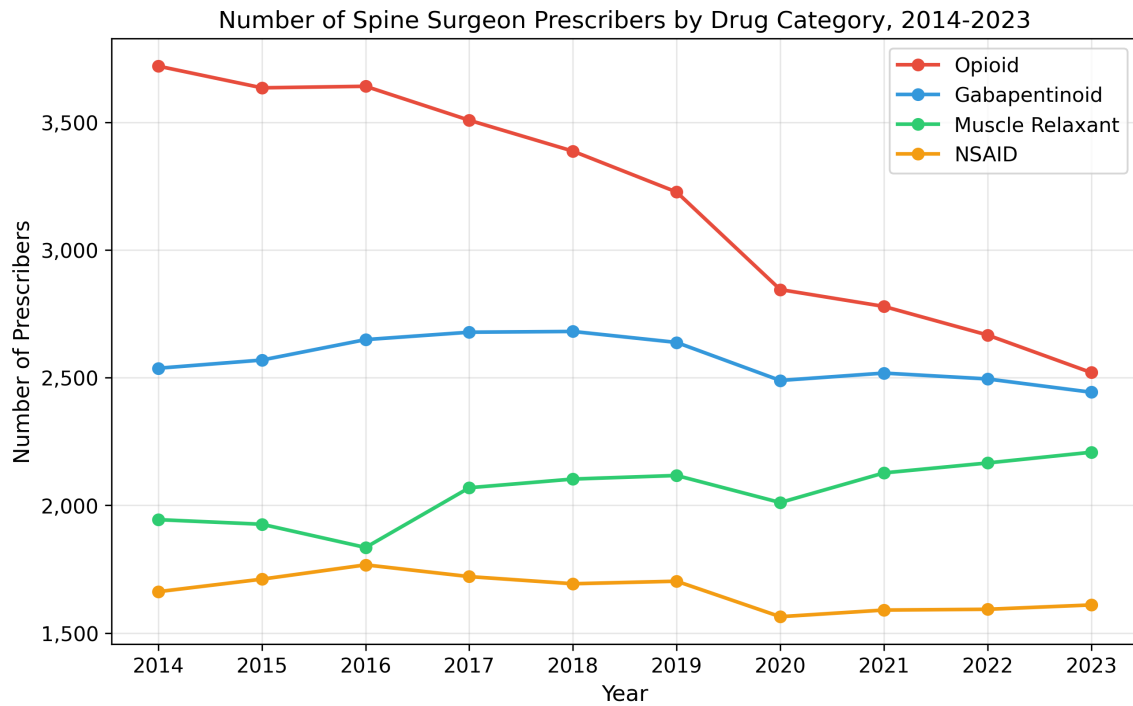


Figure 4: Average Prescription Duration (Days Supply Per Claim)

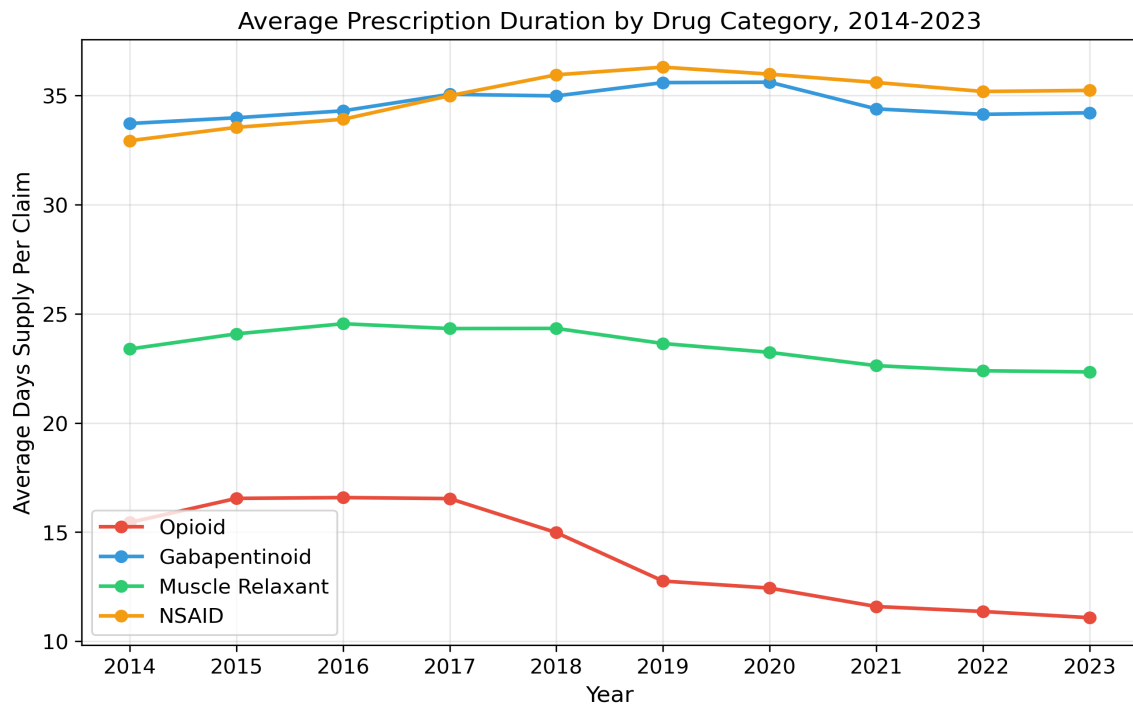


Figure 5: Normalized Prescribing Trends (2014 = 100)

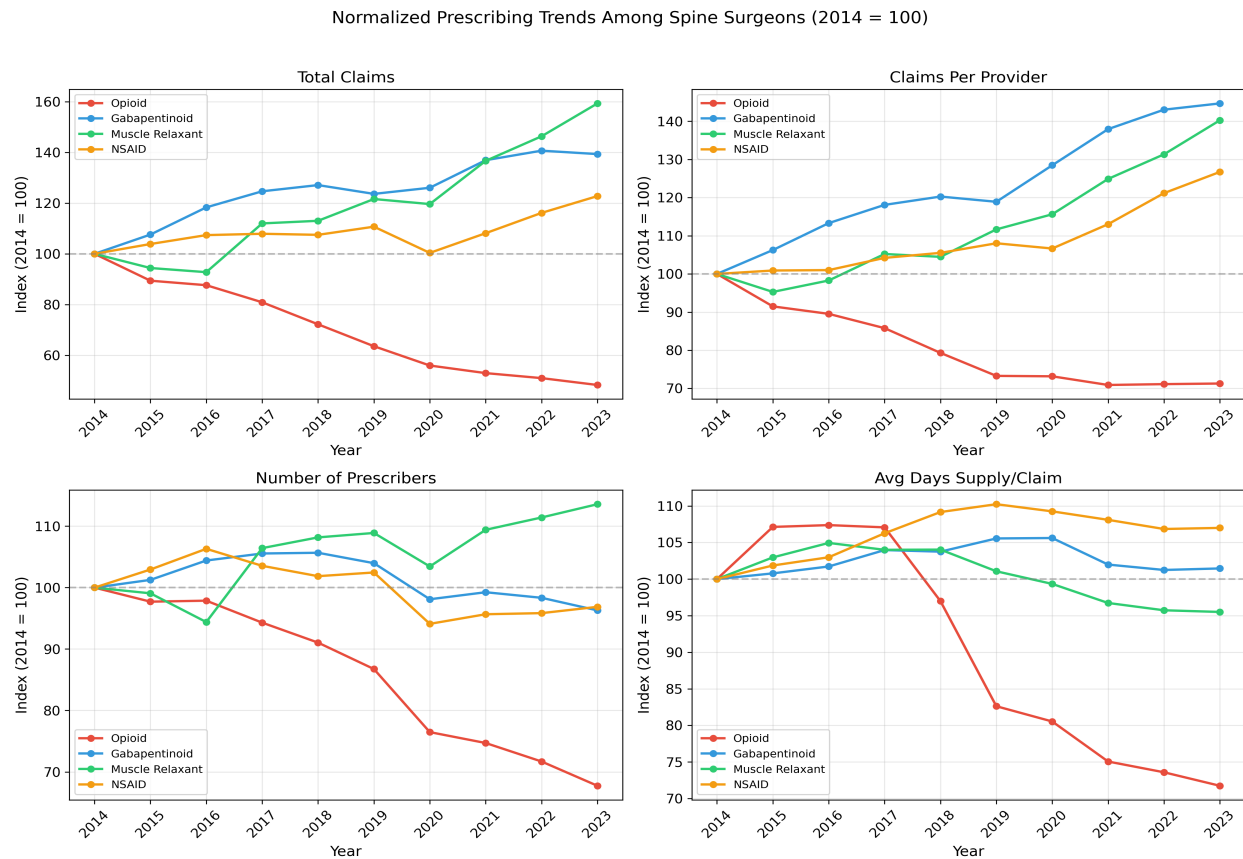


Figure 6: Total Drug Cost by Category

Total Drug Cost by Category Among Spine Surgeons, 2014-2023

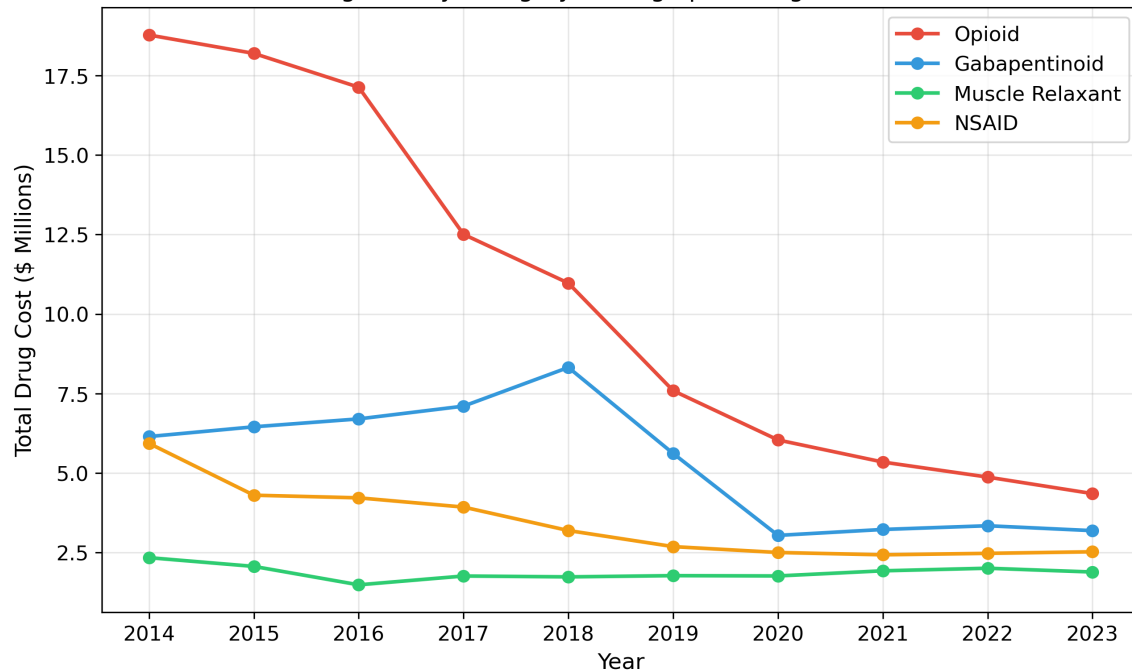


Figure 7: Distribution of Pain Management Claims by Category

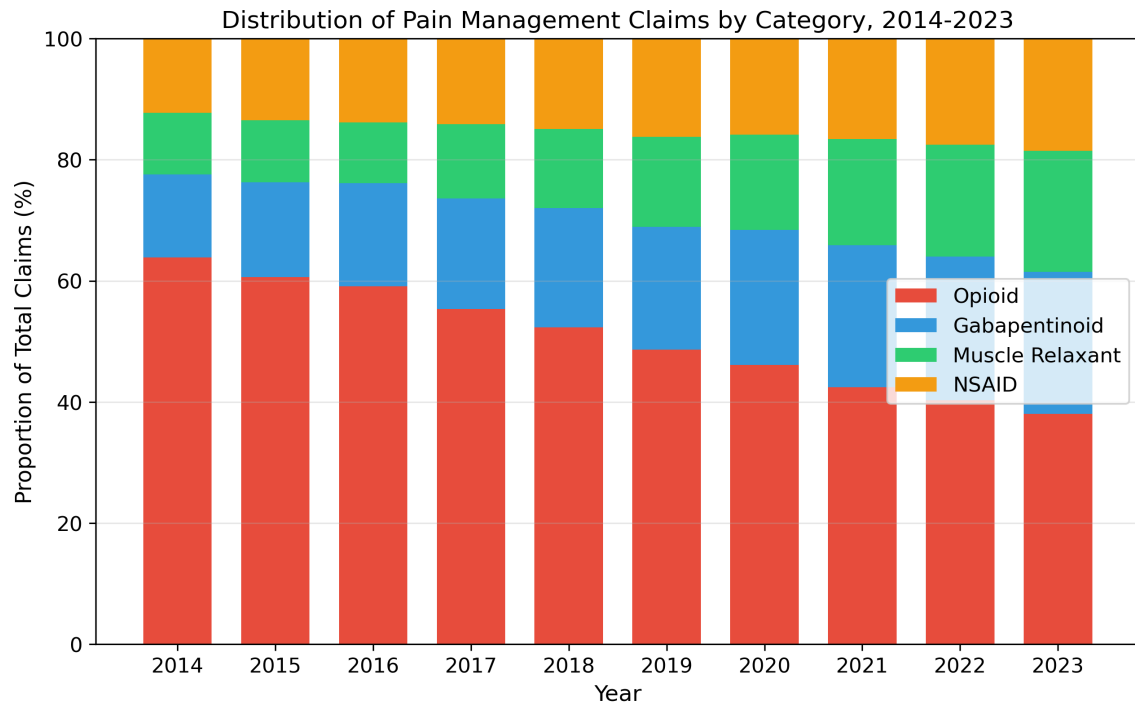


Figure 8: Percent Change in Prescribing Metrics, 2014–2023

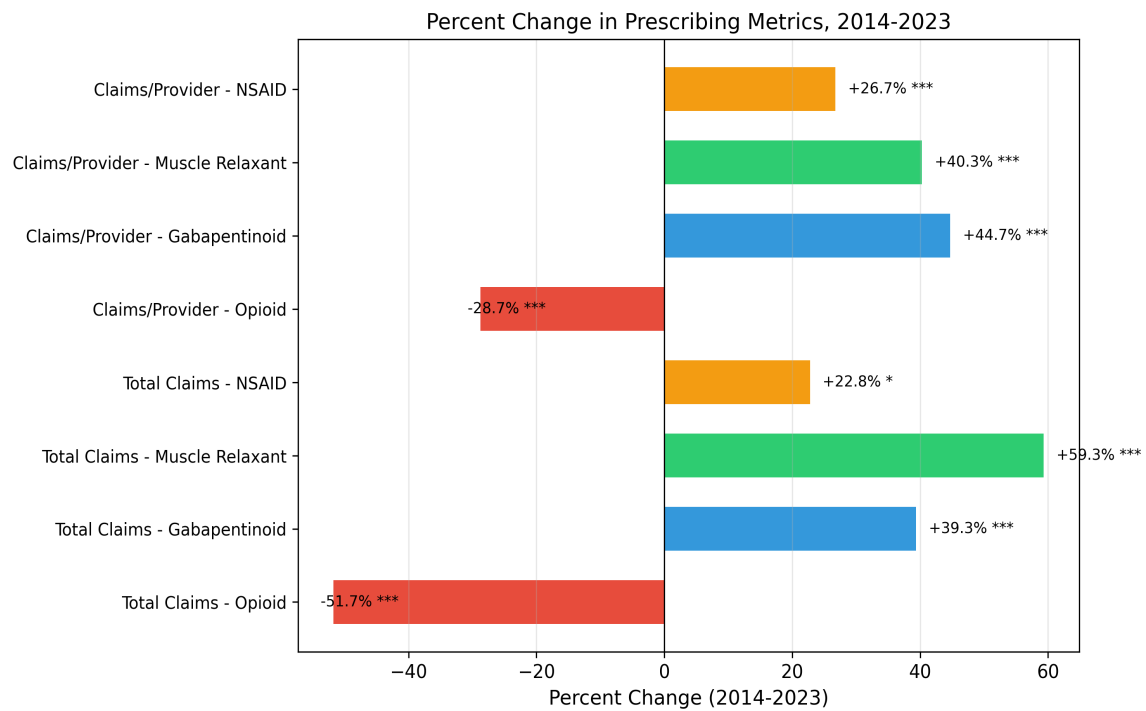


Table 8: Linear Regression Results (All 10 Years)

Metric	Slope/yr	R ²	P-value	% Change
Total Claims - Opioid	-42,526	0.969	<0.001***	-51.7%
Total Claims - Gabapentinoid	+6,236	0.891	<0.001***	+39.3%
Total Claims - Muscle Relaxant	+7,873	0.903	<0.001***	+59.3%
Total Claims - NSAID	+2,261	0.535	0.016*	+22.8%
Claims/Provider - Opioid	-6.21	0.893	<0.001***	-28.7%
Claims/Provider - Gabapentinoid	+2.94	0.964	<0.001***	+44.7%
Claims/Provider - Muscle Relaxant	+2.77	0.909	<0.001***	+40.3%
Claims/Provider - NSAID	+2.24	0.850	<0.001***	+26.7%
Prescriber Count - Opioid	-146	0.955	<0.001***	-32.3%
Avg Days Supply - Opioid	-0.70	0.838	<0.001***	-28.3%
Total Drug Cost - Opioid	-\$1.85M	0.937	<0.001***	-76.8%
Cost Per Claim - Opioid	-\$1.95	0.916	<0.001***	-52.0%
Cost Per Claim - Gabapentinoid	-\$3.54	0.786	<0.001***	-62.8%
Cost Per Claim - NSAID	-\$2.74	0.840	<0.001***	-65.3%
Spine Surgeon Count	-44	0.554	0.014*	-6.5%
Pain Rx Prescriber Count	-91	0.894	<0.001***	-18.7%

* p<0.05, ** p<0.01, *** p<0.001

Methods

Data Source: CMS Medicare Provider Utilization and Payment Data, accessed via the CMS data.cms.gov API (Socrata Open Data API). All 10 years (2014–2023) included.

Spine Surgeon Identification: Providers were identified from Medicare Part B Physician/Supplier data as those with provider type 'Neurological Surgery' or 'Orthopedic Surgery' who billed for spine-specific CPT codes (22551, 22554, 22612, 22630, 22633, 22840, 22842, 63005, 63030, 63042, 63047, 63056, 63075, 63081, 22800, 22802, 22804, 22808, 22810, 22812, among others). This procedure-confirmed approach ensures the cohort represents surgeons actively performing spine procedures.

Drug Categories: Four categories of pain management medications were analyzed: (1) **Opioids** — hydrocodone, oxycodone, tramadol, morphine, fentanyl, codeine, hydromorphone, methadone, oxymorphone, tapentadol, buprenorphine, meperidine; (2) **Gabapentinoids** — gabapentin, pregabalin; (3) **Muscle Relaxants** — cyclobenzaprine, methocarbamol, tizanidine, baclofen, metaxalone, carisoprodol, orphenadrine, dantrolene, chlorzoxazone; (4) **NSAIDs** — meloxicam, diclofenac, naproxen, celecoxib, ibuprofen, indomethacin, ketorolac, piroxicam, sulindac, etodolac, nabumetone, ketoprofen, oxaprozin, fenoprofen, flurbiprofen, mefenamic acid, tolmetin.

Prescribing Metrics: Total Medicare Part D claims, number of unique prescribers, mean claims per provider, average days supply per claim, total drug cost, cost per claim, cost per beneficiary, and total 30-day fill equivalents were calculated for each drug category per year. Specialty-level analysis stratified results by neurosurgery vs. orthopedic surgery.

Statistical Analysis: Linear regression was used to estimate temporal trends (slope per year) for each metric across all 10 years. Percent change was calculated from 2014 baseline to 2023.

Limitations: The analysis is limited to Medicare Part D claims and does not capture prescriptions filled through other payers (commercial insurance, Medicaid, cash pay, or hospital/facility dispensing). Spine surgeon identification is based on Medicare billing and may not capture all spine surgeons, particularly those who do not participate in Medicare. CMS suppresses data for providers with fewer than 11 claims for a given drug, which may undercount low-volume prescribers.